College of Engineering Pune

(An Autonomous Institute of the Govt. of Maharashtra)

YEAR : SECOND YEAR BACHELOR OF PLANNING
SEMISTER: IV

SUBJECT: Geo-informatics-I

Lecture 3

Prepared by,

Jaysing Jadhav
Adjunct Faculty, Planning section,
Dept. of Civil Engineering CoEP

DEFINITION OF GIS

- A Geographic Information System should be thought of as being much more than means of
  - **coding, storing, and retrieving** the data about the aspects of earth's surface, because these data can be
  - **accessed, transformed, and manipulated interactively** for studying **environmental process, analyzing the results for trends, or anticipating the possible results of planning decisions.**
- Geographical Information System is associated with basic terms:
  - Geography
  - Information system.

- **GEOGRAPHY** is ‘writing about the Earth’.
  - In writing about the Earth, geographers deal with the spatial relationship of land with man.
- To study the spatial relationships
  - map which is a graphical portrayal of spatial relationships and phenomena over a small segment of the Earth or the entire Earth.
- **INFORMATION SYSTEM**
  - chain of operations that consists of from planning the observation to using the observation-derived information in some decision making process.

- GIS is both
  - a **database system** with specific capabilities for spatially-referenced data
  - **set of operations** for working with the data.
- Some of the **definitions of GIS** given in different publications are

DEFINITION OF GIS

- *“A system which uses a spatial database to provide answers to queries of a geographical nature”.*

- *“A computer assisted system for the capture, storage, retrieval, analysis, and display of spatial data within a particular organization”.*

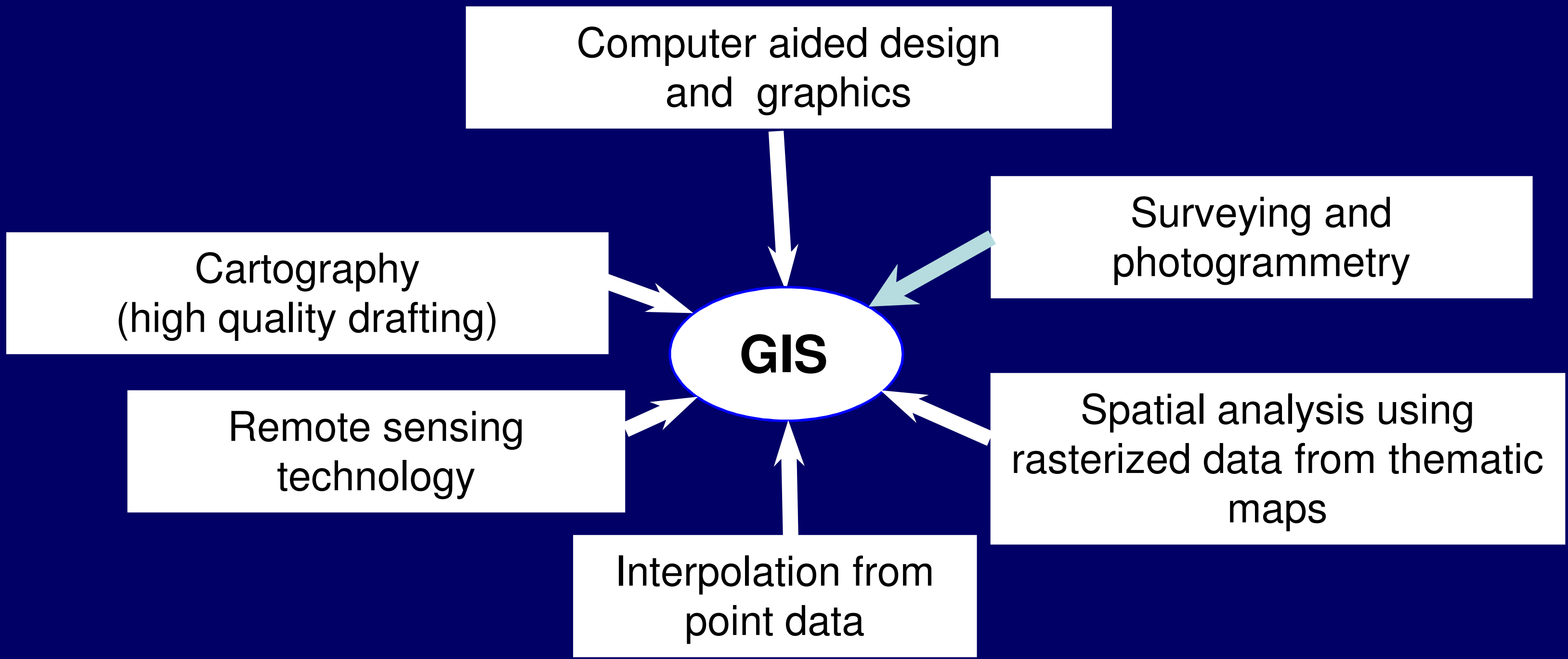
- *“A powerful set of tools for collecting storing, retrieving at will, and displaying spatial data from the real world*

COMPLETE DEFINITION

- *"An organized collection of*
  - *computer hardware,*
  - *software,*
  - *geographical data, and*
  - *personnel*

  designed to efficiently capture, store, update, manipulate, analyze, and display all forms of geographically referenced information" (ESRI).

- A GIS is also the result of linking parallel developments in many separate spatial data processing.



Linking of several related fields though GIS

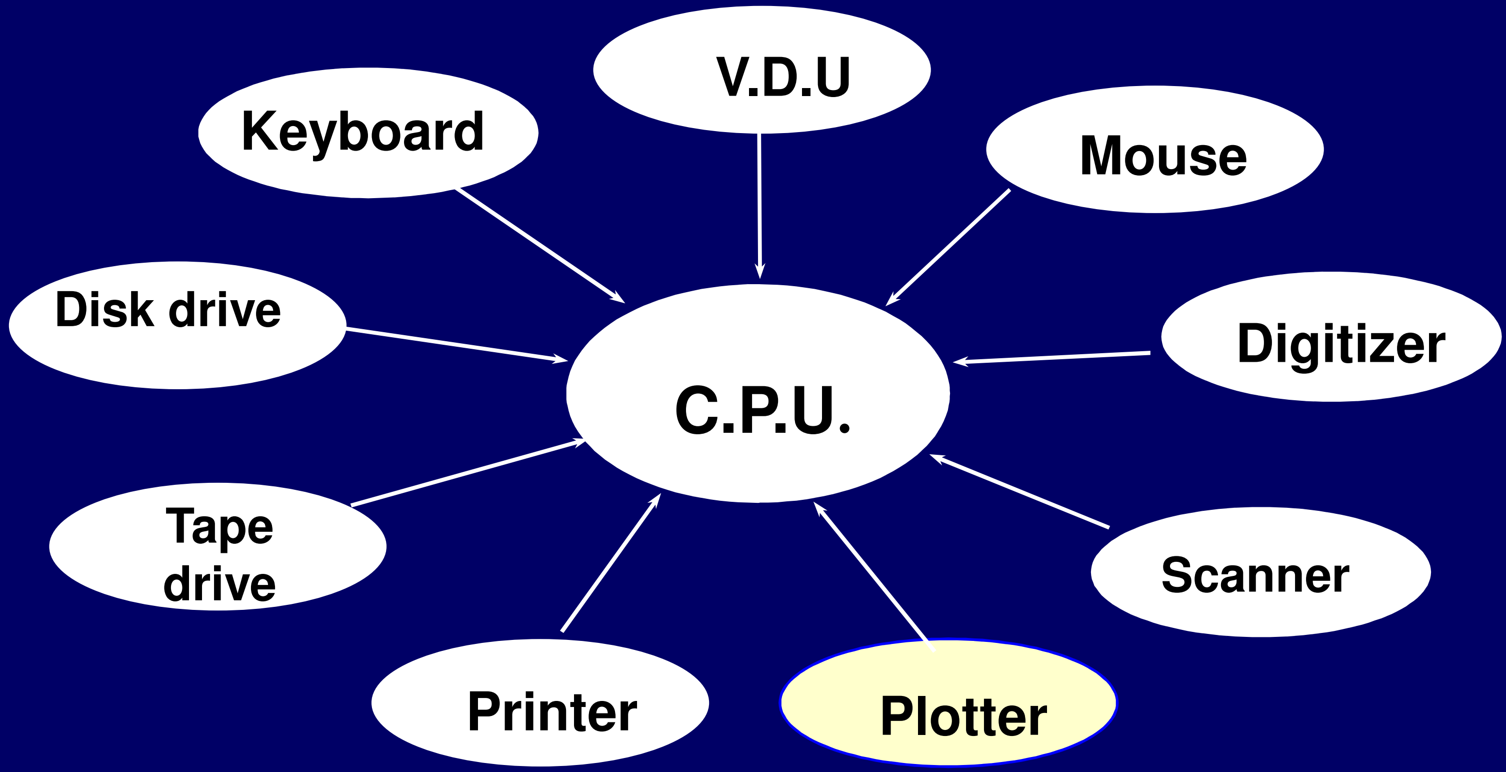
COMPONENTS OF GIS

- The components of GIS are:

(*i*) Computer system (hardware & operating system),
(*ii*) Software,
(*iii*) Spatial data,
(*iv*) Data management and analysis procedures,
(*v*) Personnel to operate the GIS.

Hardware Components

- The hardware components of a GIS comprise of a
  - Central Processing Unit (C.P.U.),
  - disk drive,
  - tape drive,
  - digitizer,
  - plotter, and
  - visual display unit (V.D.U.)
- The disk drive and tape drive are basically data storage devices.
- The tape can be used for communicating with other systems.



Major hardware components of a GIS

- A digitizer and scanner are input devices to convert graphics into digital data.
- The Visual Display Unit along with a keyboard or mouse is required to interact with the computer.
- The printer is required to get hardcopy of the reports, tables, charts, etc where as the plotter, an output device, converts the digital data into the graphical form.
- The Central Processing Unit of a computer interacts with various hardware components, and performs computations and analysis.

SPATIAL DATA

- Spatial data are characterized by information about position, connections with other features and details of non-spatial characteristics.
- All GIS software are designed to handle spatial data.
- Spatial data require spatial referencing using a suitable geographic referencing system which should be flexible and lasting, since a GIS may be intended to last many years.

TYPES OF OUTPUT PRODUCTS

a) Thematic maps
b) Chloropleth maps
c) Proximal or dasymetric maps
d) Contour maps
e) Dot maps
f) Line maps
g) Land form maps
h) Animated maps
i) Non-map graphics

- **Thematic maps**
  - spatial variations of a single phenomenon (*e.g.*, population) or
  - relationship between phenomena (*e.g.*, different classes of land cover).

- Choropleth maps
  - relative magnitudes of continuous variables as they occur within the boundaries of unit areas (*e.g.*, average annual per capita income).

- In these maps, different tones, colours, and shading patterns are used to convey the variations in different areas.

- Proximal or dasymetric maps
  - location and magnitude of areas exhibiting relative uniformity (*e.g.*, land cover classes).
- Different colours and shading patterns are used to describe differences in the thematic values.

- Contour maps
  - quantities by lines of equal value to emphasize gradients among the values.

  Contour lines may be used to indicate variation in topography of a region, high and low pressure regions

- Dot maps
  - spatial distribution of features by varying numbers of uniform dots (*e.g.*, population)

- Line maps
  - direction and magnitude of potential or actual flow

    (*e.g.*, to show sources and destinations as well as the volume of product transported from one state to others).

- Land form maps depict the earth's surface as it were viewed from an oblique aerial point view.

- Animated maps are generally used to display sequences through time (*e.g.*, growth of a city as its population and area increase through time).

GIS DATA TYPES

Geographic data consists of

a) *spatial data*

b) *non-spatial* data.

SPATIAL DATA

- It gives information about
  - the geometrical orientation,
  - shape and size of a feature, and
  - its relative position with respect to the position of other features.

- Spatial data is described by its *x* and *y* coordinates.
- The spatial data is normally available in analog form as maps but now the maps are also available directly in digital format.
- In GIS, both types of the spatial data are handled differently.
- Normally the spatial and non-spatial data are stored separately in a GIS, and links are established between the two at the time of processing and analysis.

NON-SPATIAL DATA

- The non-spatial data, also known as *attribute data*, are information about various attributes like length, area, population, acreage, etc.
- The non-spatial data describe the attributes of a point, along a line, or in a polygon.
- In other words they describe what is at a point (*e.g.*, a hospital), along a line (*e.g*., a canal), or in a polygon (*e.g*., a forest).
- The attributes of a soil category may be depth of soil, texture, type of erosion, or permeability.
- The non-spatial data, mostly available in tabular form, are also converted into digital format for use in GIS.

SPATIAL DATA STRUCTURES

- Data structures provide the information that the computer requires to reconstruct the spatial data model in digital form.
- There are many different data structures in use in GIS.
- In general, data structures can be classified according to the type of data they use,

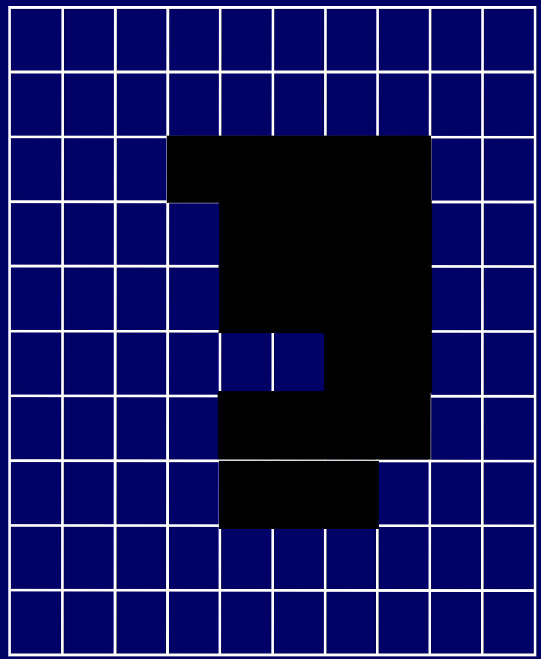
a) raster

b) vector data.

RASTER DATA STRUCTURES

- Coding of raster data of the entity model, *i.e.*, an image is done by having mirror image of the equivalent row of numbers in the file structure, in the cells in each line of the image, where the entity is present.

RASTER DATA STRUCTURE



(a) Entity model

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1	0	0
0	0	0	0	1	1	1	1	0	0
0	0	0	0	1	1	1	1	0	0
0	0	0	0	0	1	1	1	0	0
0	0	0	0	1	1	1	1	0	0
0	0	0	0	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

(b) Cell values

10,10,1

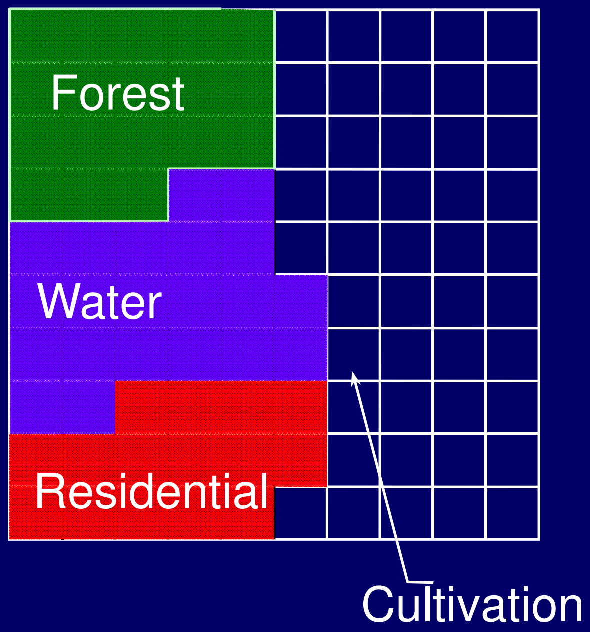
0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 1, 1, 1, 1, 1, 0,
0, 0, 0, 0, 1, 1, 1, 1, 0,
0, 0, 0, 0, 1, 1, 1, 1, 0,
0, 0, 0, 0, 0, 1, 1, 1, 0,
0, 0, 0, 0, 1, 1, 1, 1, 0,
0, 0, 0, 0, 1, 1, 1, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0,

(c) File structure

- The first line in the raster file is basically a header that gives information regarding the size of the file and the maximum value that a cell may have.
- First two values indicate that there are 10 rows and each row has 10 pixels/columns, while the value 1 indicates the maximum value that a cell may have.
- Thus in this example, the value of 0 represent cells having no entity value while 1 indicates the presence of an entity.

- In raster data structure, different spatial features, such as building, water bodies, contours, roads, maybe stored as separate data layers, *i.e.*, these features are stored in four separate data files, each representing a different layer of spatial data.
- However, if the entities do not occupy the same geographic location (or cell in the raster model) then it is possible to store them all in a single layer with an entity code given to each cell.
- This code indicates that which entity is present in a cell.
- An example of this case is shown in Fig.

Feature Coding of Cells



2	2	2	2	2	1	1	1	1	1
2	2	2	2	2	1	1	1	1	1
2	2	2	2	2	1	1	1	1	1
2	2	2	3	3	1	1	1	1	1
3	3	3	3	3	1	1	1	1	1
3	3	3	3	3	3	1	1	1	1
3	3	3	3	3	3	1	1	1	1
3	3	4	4	4	4	1	1	1	1
4	4	4	4	4	4	1	1	1	1
4	4	4	4	4	1	1	1	1	1

10,10,4,
2, 2, 2, 2, 2, 1, 1, 1, 1,
2, 2, 2, 2, 2, 1, 1, 1, 1,
2, 2, 2, 2, 2, 1, 1, 1, 1,
2, 2, 2, 3, 3, 1, 1, 1, 1,
3, 3, 3, 3, 3, 1, 1, 1, 1,
3, 3, 3, 3, 3, 3, 1, 1, 1,
3, 3, 3, 3, 3, 3, 1, 1, 1,
3, 3, 4, 4, 4, 4, 1, 1, 1,
4, 4, 4, 4, 4, 4, 1, 1, 1,
4, 4, 4, 4, 4, 1, 1, 1, 1,

(a) Entity model (b) Cell values (c) File structure

VECTOR DATA STRUCTURE

- The entities point, lines, and areas can be defined by a coordinate system.
- One of the simplest vector data structure to represent a geographical image in the computer is a file containing (x, y) coordinates that represent the location of individual point features or the points used to construct lines and areas.

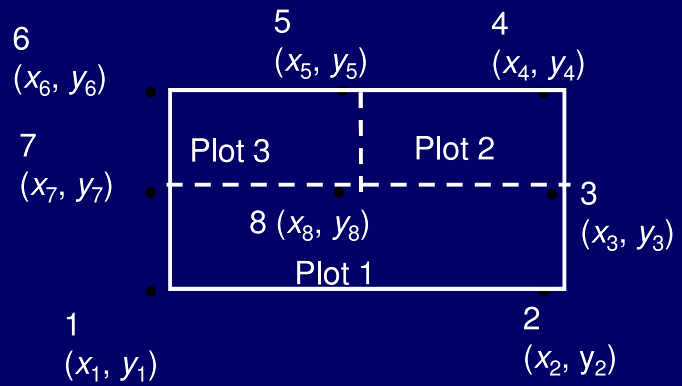
- Fig. a shows a simple vector data structure using point coordinates to describe a parcel of land.

"Spaghetti file"	
X	**Y**
X1	Y1
X2	Y2
X3	Y3
X4	Y4
X1	Y1

(a) Simple data structure

- The limitation of this approach emerges in dealing with more complex spatial entities.

- For example, when the parcel of land shown is divided into plots 1, 2, and 3, the individual plots become a number of adjacent polygons.



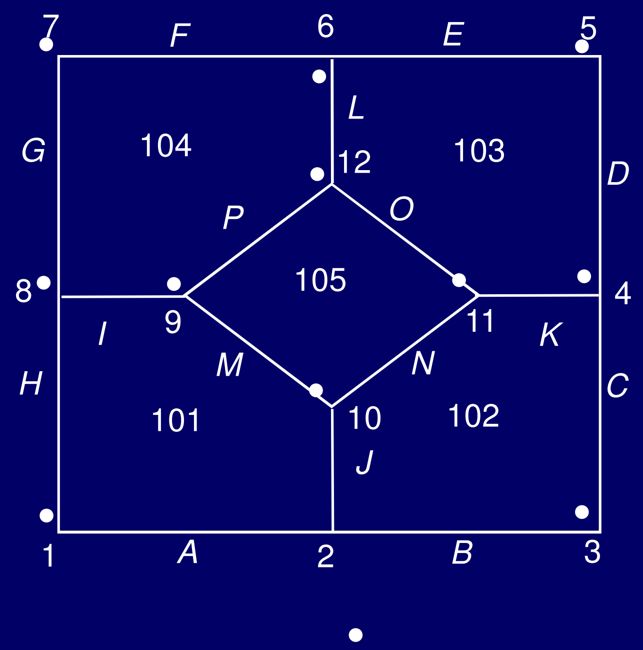
- If the simple vector data structure as used in is used, there will be duplication of data to define each entity individually, and the problem becomes serious when there are a large number of entities.

- To address this problem, sharing of common coordinates for adjacent polygons is done point reference numbers given sequentially indicating that which points are associated with which points.
- This is known as a *point directory* (Fig.)

Point dictionary file	
1	**x1, y1**
2	**x2,y2**
3	**X3, y3**
....	**.....**
......	**......**
8	**X8, y8**

"Polygon" file	
ID	**Point**
Plot 1	1,2,3,7
Plot 2	3,4,5,8
Plot 3	5,6,7,8

- The problem gets complicated further for an example shown in Fig.
- There are four areas for stalls from different regions and in the centre there is an information centre.



- While the simple vector file structure can recreate the image of the whole area, it cannot inform the computer that the information centre lies within the larger area containing the stalls.
- For the representation of points, lines, and polygons shown in Fig. and the entities adjacent to them, a **topological data structure** that contains this information is required.

Topological Structuring

Node file

ID	*x* (m)	*y* (m)
1	0	0
2	10	0
3	20	0
4	20	10
.		
.		
11	15	10
12	10	15

Segment file

ID	Start node	End node	Left polygon	Right polygon	Length (m)
A	1	2	101	Outside	10
C	2	3	102	Outside	10
C	3	4	102	Outside	10
D	4	5	103	Outside	10
.	.	.	.	.	.
.	.	.	.	.	.
O	11	12	105	103	7.07
P	12	9	105	104	7.07

Polygon file

ID	Segment list
101	*A, J, M, I, H*
102	*B, C, K, N, J*
103	*D, E, L, O, K*
104	*F, G, I, P, L*
105	*M, N, O, P*

Polygon attribute file

I D	VAR 1 (Name)	VAR2 (Area in m^2)
101	Punjab stall	87.5
102	Rajasthan stall	87.5
103	Maharashtra stall	87.5
104	Kerala stall	87.5
105	W. Bengal stall	50

- There are numerous ways of providing topological structure, the example given in overleaf illustrates the basic principles of topological data structure.
- Topological data structure represents **connectivity** between entities and not their physical shape.
- Table presents the basic requirements of the entities of point, line, and area to have topology.

Topological requirements of basic spatial entities

Entity	Requirement
Point	Geographical reference to locate it with respect to other spatial entities.
Line	Ordered set of points (known as an arc, segment or chain) with defined start and end points (nodes) which also give the line direction.
Area	Data about the points and lines used in construction of the area, and how these are connected to define the boundary.

Basic Requirements of a Topological Data Structure

(*i*) No node or line segment should be duplicated,

(*ii*) Line segments and node can be referenced to more than one polygon,

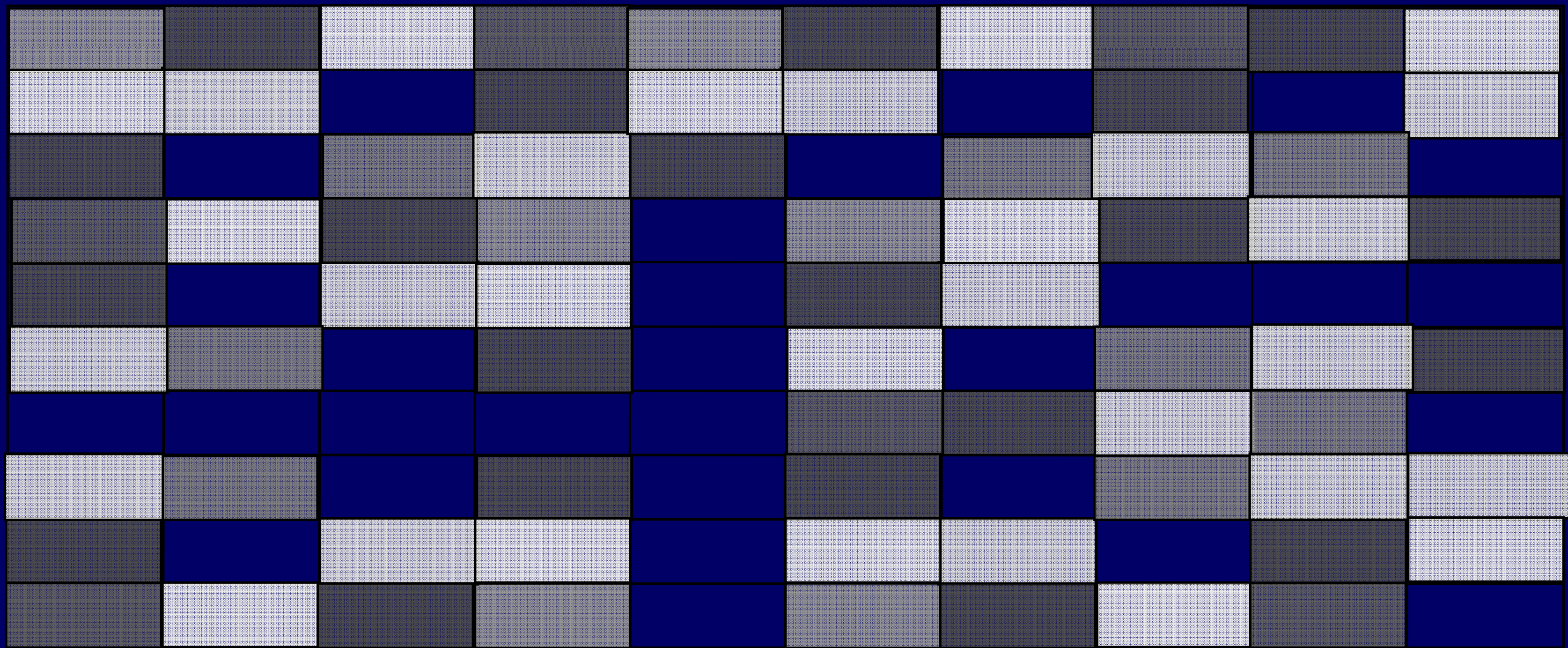
(*iii*) All polygons should have unique identifiers, and

(*iv*) Island and hole polygons can be adequately represented.

RASTER APPROACH

In raster GIS, a DTM is represented by a grid in which each cell contains a single value of the height or the terrain covered by that cell.

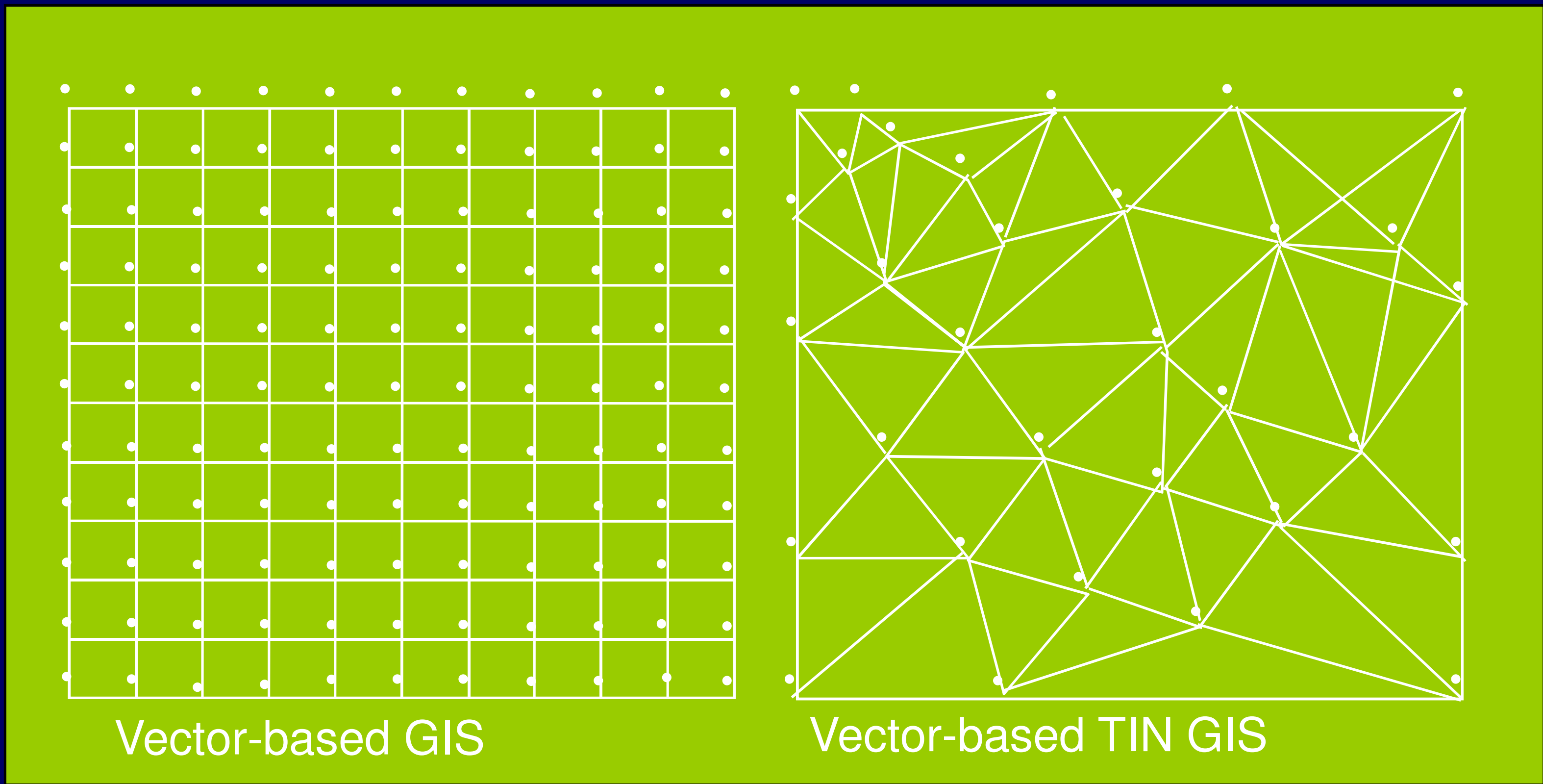
The accuracy of the terrain modeling by this approach depends on the spacing (resolution) of the grid and complexity of the terrain surface.



Raster-based GIS

VECTOR APPROACH

- A vector DTM is like a raster DTM with the difference that the vector DTM has regularly spaced set of spot heights (Fig.).



VECTOR APPROACH

- In vector GIS, to represent the terrain surface with irregular height data, the common practice is to use **TIN**, which creates a mosaic of irregular triangles by joining the points where spot heights are known by straight lines.
- The vertices of the triangles in a TIN represent the features such as peaks, depressions, and passes, and the sides represent ridges and valleys whereas the surface of the individual triangle provides area, gradient (slope), and orientation (aspect).
- Storing these values as TIN attributes helps in further analysis.

  TIN model has the following advantages:

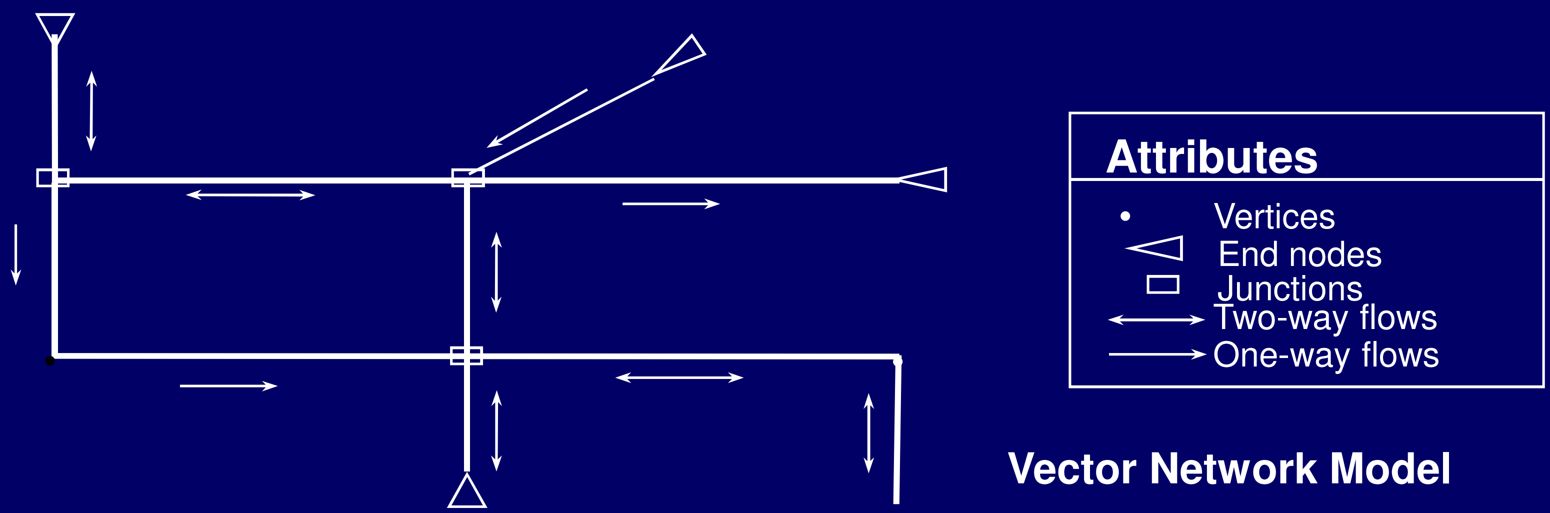
(*i*) Efficiency of data storage

(*ii*) Fewer points required to represent flatter areas.

MODELLING NETWORKS

- A set of **interconnected linear features** such as roads, railways, telephone lines, through which goods and people are transported or along which communication of information is processed, is said to be a ***network***.
- Since network models in GIS are abstract representations, the use of vector data model and raster data model is not suited for network analysis.

- Fig. shows a vector network model made up of the some arc (line segments) and node elements as any other vector data model but with the addition of special attributes.



Vector Network Model

Definition of Attributes in Networks

Attributes	Features represented
Arc	**Network links:** Roads, railways, power lines, cables and pipelines of utilities, rivers, streams.
Node	**End points of network links:** Stops, Centres, junctions in transport networks, confluences in stream networks, switches and values in utility network.
Stop	**Locations may be visited:** Bus stop, pick-up and delivery points in a delivery system.
Centre	**Discrete locations in a network:** Shopping centres, airports, hospitals, schools, hospitals, cities (at smaller scale)
Turn	**Transition from network link to network link at a network node:** Turns across oncoming traffic on a road network take longer than turns down shipways, whereas turns that go against the flow of traffic on one-way roads are prohibited altogether.

DATABASE STRUCTURES

- Three fundamental ways of organizing information that also reflect the logical models used to model real world structures as below:

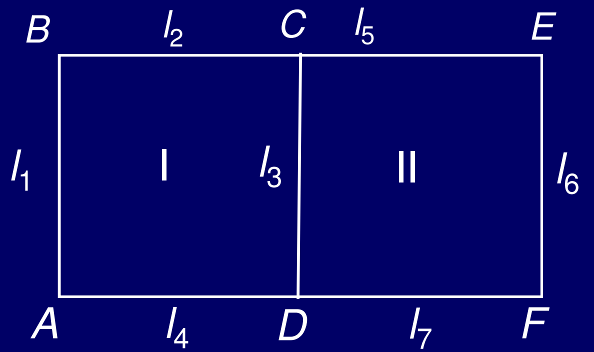
(*i*) Hierarchical database structure

(*ii*) Network database structure

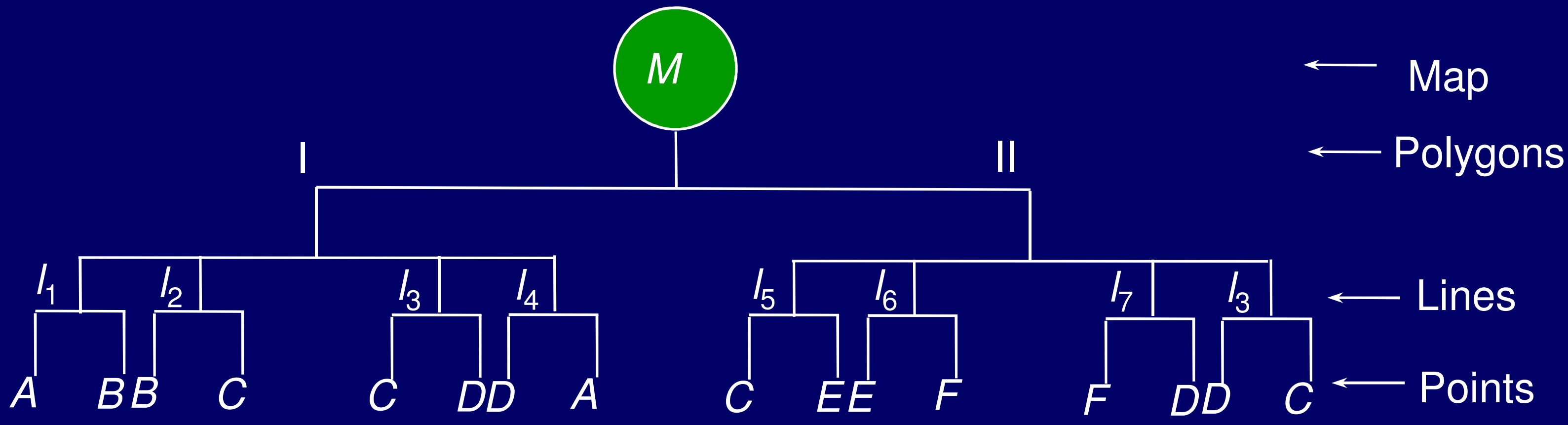
(*iii*) Relational database structure.

Hierarchical Database Structure

- A data may have a multi-layered data with a direct relationship between each layer, similar to a tree like structure.



(b) A simple polygon map



(b) Hierarchical data structure

- The relationship between two successive layers is known as *parent-child* or *one to many* relationship.
- For such type of data, hierarchical database provides a quick and convenient means of data access.
- Here each part of the hierarchy can be reached using a *key* or criterion, and that there is a good correlation between key and associated attributes.

ADVANTAGES & DISADVANTAGES

- The main advantage of hierarchical database is that it is simple and provides easy access through keys defining the hierarchy.
- Further, it is easy to expand by adding more attribute and formulating new decision rules.
- The success of data retrieval in a hierarchical database depends upon the prior knowledge of structure of all possible queries.

- One of the biggest disadvantages of hierarchical database is the repetitive data.

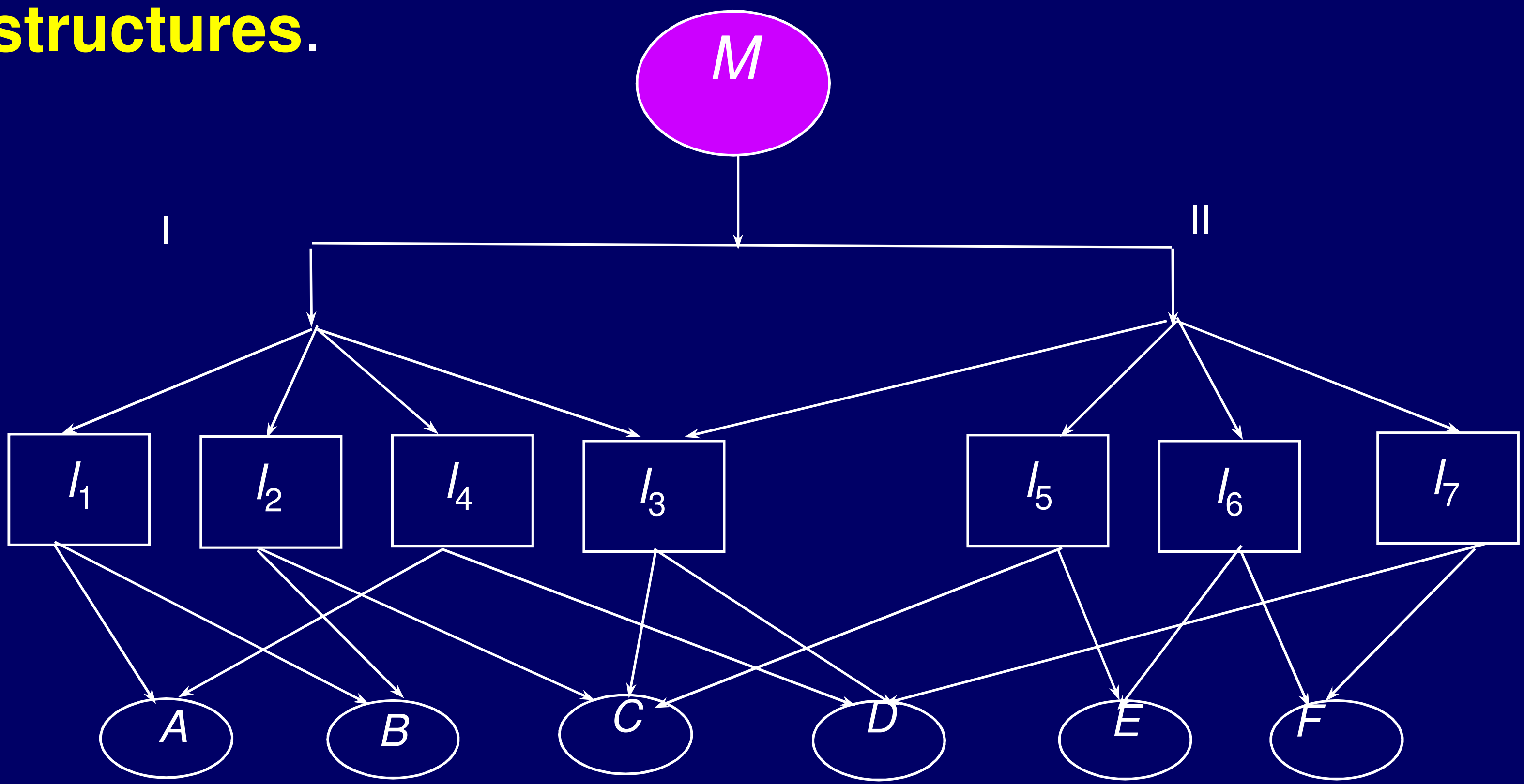
- Referring to Fig., it can be seen that each pair of points is repeated twice and that for line *l*3, the coordinates *c* and *d* are repeated four times.

- This is a simple wastage and causes large redundancy of data in case of large databases.
- In hierarchical structures, the access within the database is restricted to paths up and down the hierarchy levels.

Network Data Structures

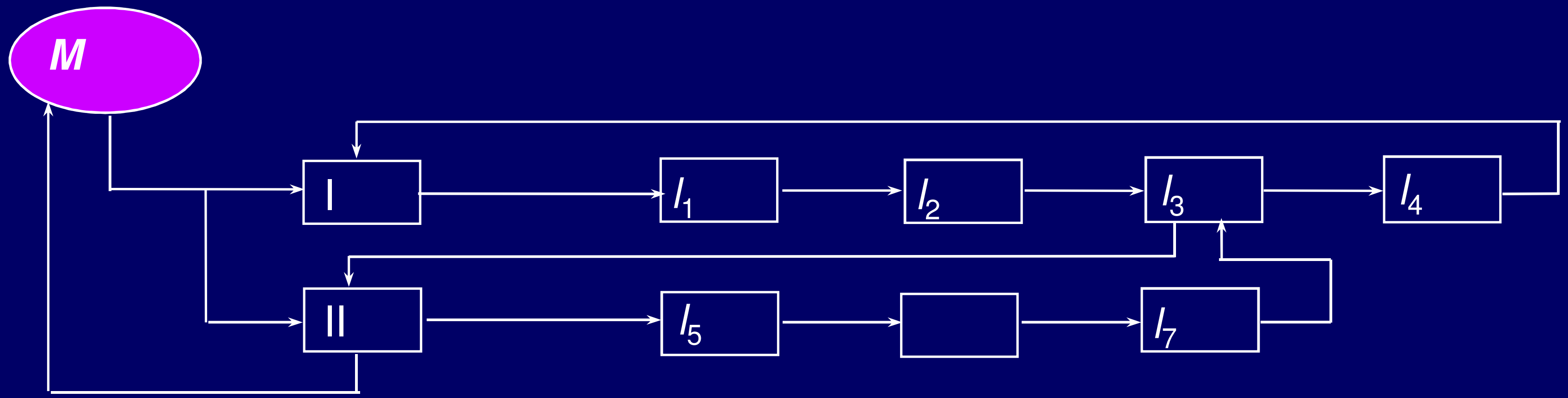
- Many a times, a single entity may have many attributes, and **each attribute is linked to many entities**.
- To accommodate these relationships, each piece of data can be associated with an explicit computer structure called ***pointer*** which directs it to all the other pieces of data to which it relates (Fig.).

- Here rather than being restricted to a branching tree structure, each individual data is linked directly without the existence of a parent-child relationship.
- Such a data structure is called **Network data structures**.



RING POINTER STRUCTURES

- Sometimes in order to reduce both redundancy and linkage, a compact network structure known as ***ring pointer structures*** are used, so that each entity appears once.
- In this Fig., each point is having many linkages or point. Fig. shows the ring pointer structure, where the flow is simplified.



Ring pointer structure

- Network structures are useful when the relations or linkages are specified before hand.
- They avoid data redundancy and make good use of available data.
- Hence, they allow greater flexibility of search than hierarchical structures.

- The disadvantage of network structure is that for large databases, the number of pointers can become large, and can become a substantial part of the database.
- Further, these pointers have to be maintained every time a change to the database is made.
- Building and maintenance of pointer structures may be a considerable overhead for the database system.

RELATIONAL DATABASE STRUCTURE

- In relational database structure, data are organized in a series of two-dimensional tables, each of which contains records for one entity.
- These tables are linked by common data known as *key*.
- It is possible to make query on individual or a group of tables.

LAND TITLE Table (Attribute set-1)

Number	LID Area	Date of Registration	Area
001	03420	16.10.1986	Civil Lines
002	19286	05.08.1972	Ram Nagar
003	06072	23.11.1996	Rajputana
004	30123	02.11.1969	Rekabgan g

Attribute

PARCEL Table (Attribute set-2)

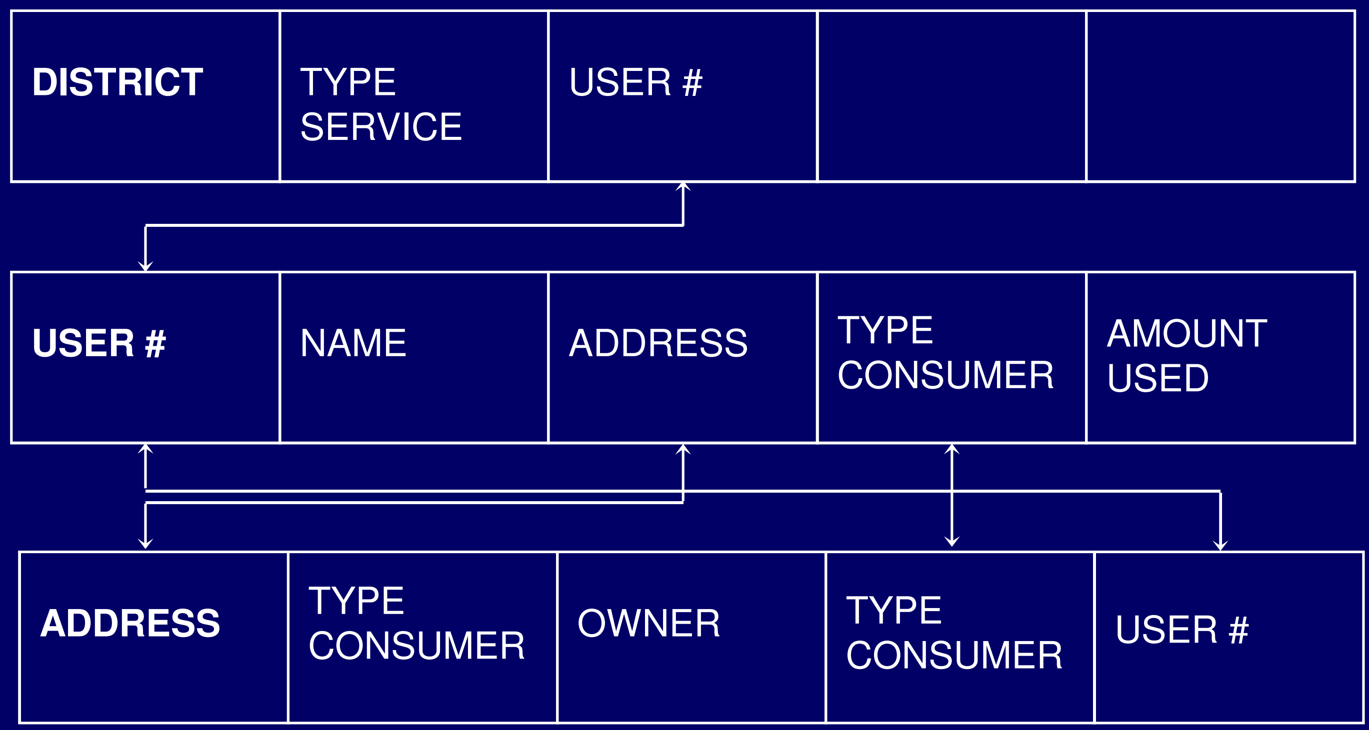
LID	Size of parcel (m^2)	Category	Registration #
03420	150	*B*	86-12032
19286	185	*A*	72-3306
06072	200	*B*	96-1985
30123	160	*C*	69-2066

Tuples

OWNER table (Attribute set-3)

LID	Name	Postal address	Phone #
03420	A. Chandra	213, Saraswati Kunj, Rke	271519
19286	S. Ghosh	215, Niti Nagar, Lucknow	275329
06072	P. Garg	97, Vigyan Kunj,Meerut	275060
30123	P. Bhargava	206, Amod Kunj, Dehradun	276011

- Each table contains data for one entity.
- In the first table, the entity is ‘landtitle’.
- In the second and third tables, the entities are ‘parcel’ and ‘owner’.
- The data are organized into rows and columns, with columns containing the attributes of the entity.
- Each column has a distinctive name, and here every entry in a single column must be drawn from the same domain.



(a) Example of a relational database for utility consumer management

DISADVANTAGES

- Some of the disadvantages of relational database structure are that

(i) It can not handle geographical concepts such as ‘connected to’, ‘near to’, or ‘far from’,

(*ii*) It always produces some data redundancy, and can be slow and difficult to implement,

(*iii*) It is difficult to handle complex objects such as found in GIS as there is a limited range of data types and difficulties with the handling of time.

Referance: Notes on GIS & Data structure by Prof. S. K. Ghosh